AI COMPUTING CONTINUUM




# AI Computing Continuum

## The Open Adapter Card - Light

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# Presenters



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# AICC Challenges to Address

| | Cloud<br>Public | Distributed Datacenter (DC) –<br>Private or Public | Access<br>Semi-Private | Edge / On-Prem<br>Private |
|---|---|---|---|---|
| **Location** | | | | |
| **Customer Type** | Hyperscale / Large Enterprise | Small/Med/Large Enterprise, NeoCloud, CoLo | CoSPs, CoLo, PoP | Retail Outlets, Quick Serve, Industrial, Transportation, etc. |

Rack Scale Architecture

Different servers with different management and security




Nothing in between

4x GPU with no Scale Up
1.5 - 5KW



Sovereign AI
Need for isolation
(no multi-tenancy required)

1x GPU no Scale Up
0.8KW - 2KW
Short Depth



Ruggedized
Liquid Cooled



Limited thermal & power delivery capacity

Some Deployments

8x GPU Scale Up
8 - 15KW

Hyperscale Adjacent



OAC targeted

1x to 4x GPU, no Scale Up
3KW - 5KW
Full Length



OAC - Lite

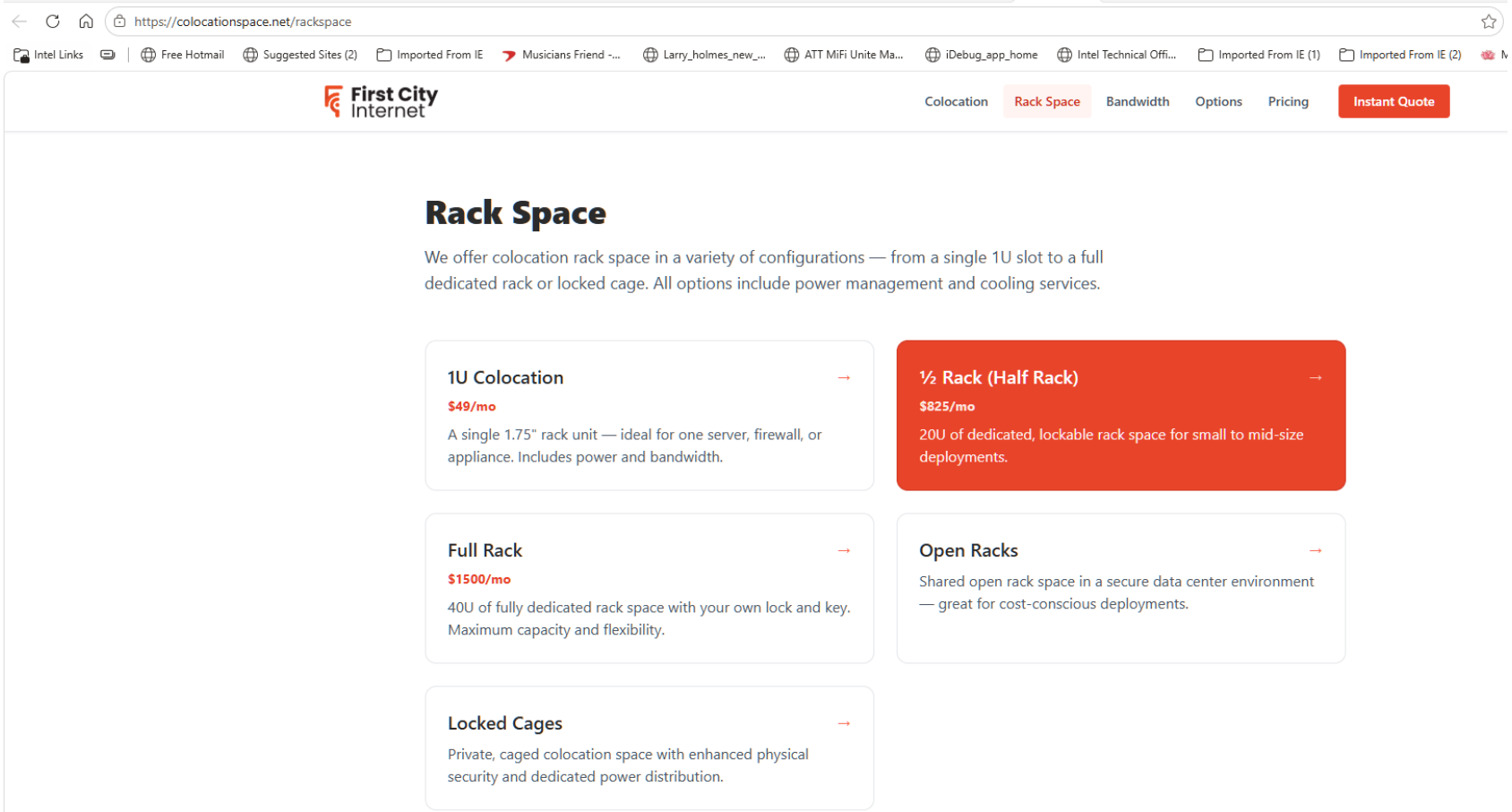
1x GPU no Scale Up
0.8KW - 3KW
Short Depth



Security Critical

# Open Adapter Card Light Challenges

## CoLos and PoPs Generally Lease Rack Space and are usually 19” cabinets




## Brownfield (Legacy) Deployments



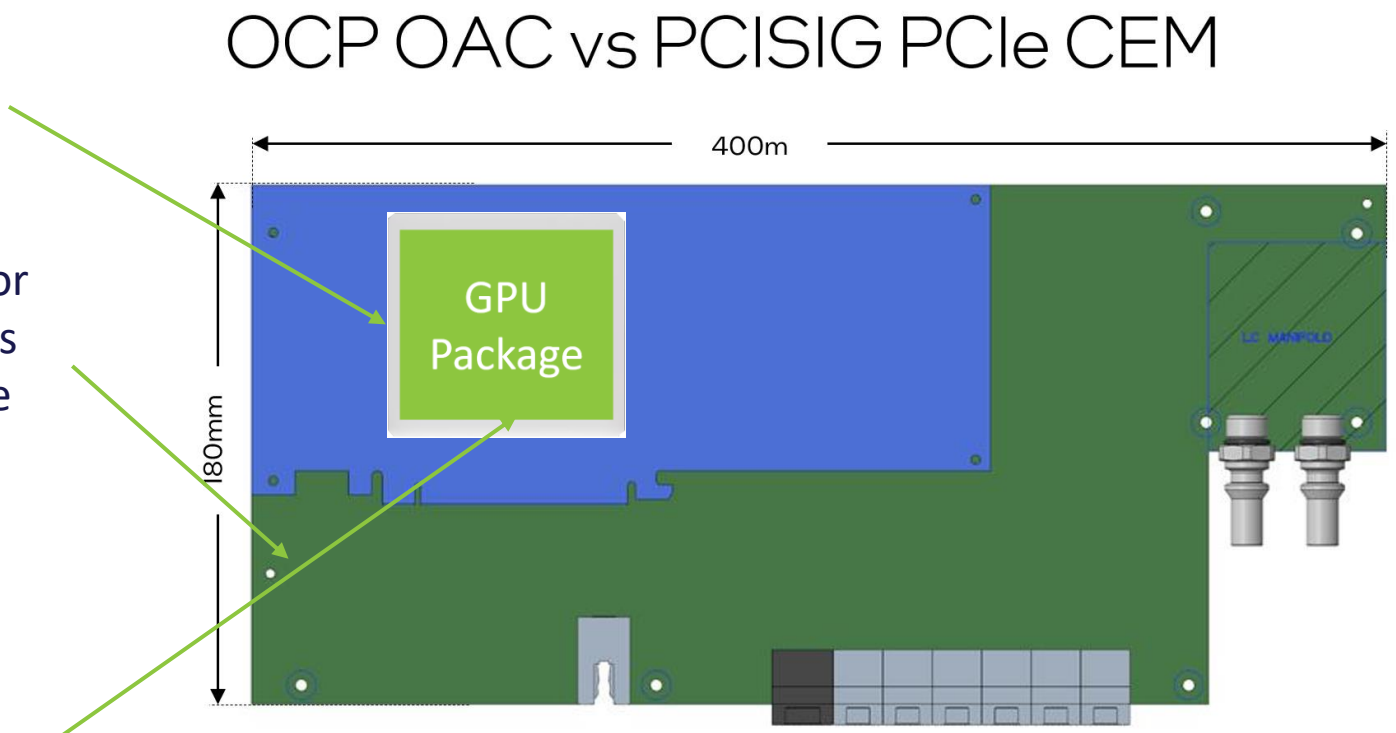
- Too Heavy
- Too Much Power
- Too Hard to cool
- Not Adjacent
- Rack & Stack (Pay-as-you-grow)

# Are OAC and PCISIG PCIe CEM Good Enough?

GPU and Switch sizes are large

OAC is very large for 1U and 2U systems but very complete functionally

PCIe CEM offers no native coherent (Scale-Up) link connectivity other than I/O Panel & future connector speed concerns

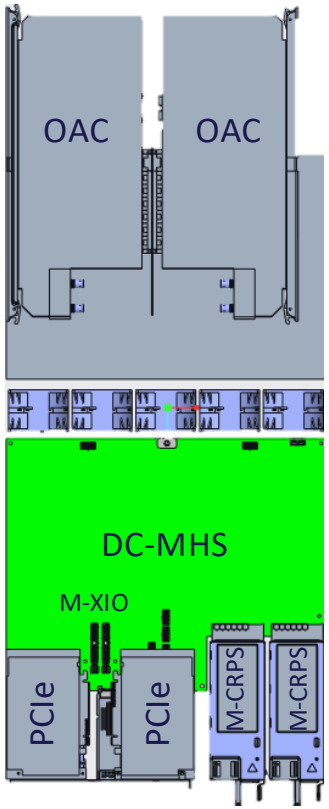


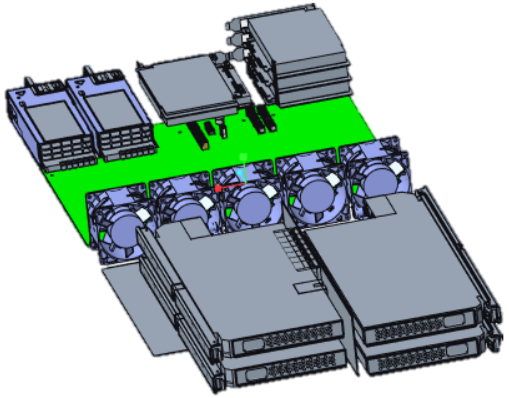

- Scale is much different, up to 120KW type racks (mostly below 50KW) and 600W Air-cooled GPU cards maximum
- 19” and 21” deployments allowing different types of scale due to legacy deployments
- GPUs are up to 1000W/GPU versus 5000W in hyperscale but below 500W in hyperscale adjacent environments
- Design optimized for air but can be liquid cooled

Are current solutions sufficient or is there something else needed?

# OAC Fit For Hyperscale Adjacent Systems

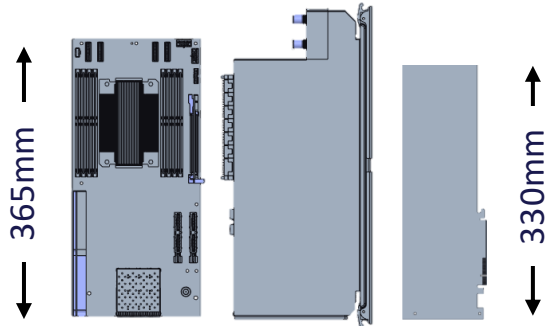
1000mm Enterprise Server



4 Node Scale-Up using OAC in 2U Rack Mount and AI Native MHS modular design

Common Add-In Card (AIC) Types

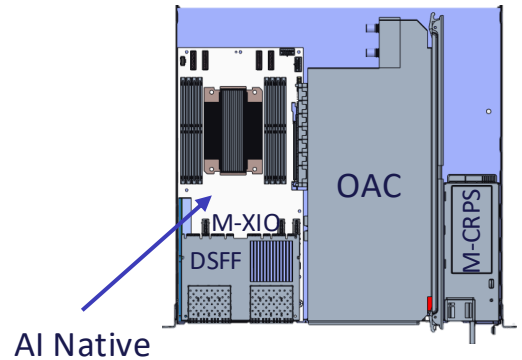



AI Native Edge-MHS 168mm wide

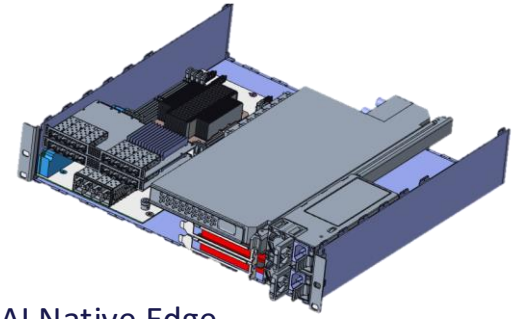
Open Adapter Card

PCIe FHFL

430mm Edge Server



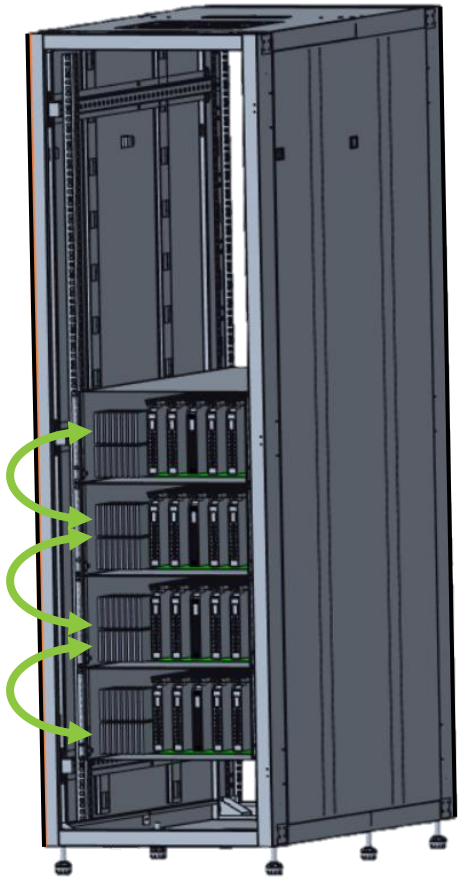

AI Native Edge-MHS



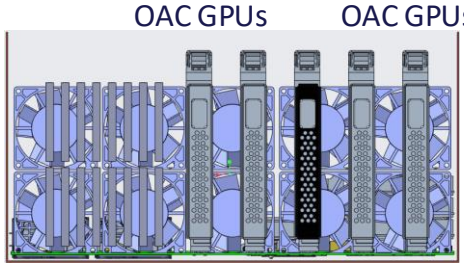
430mm AI Native Edge Solution using OAC in 2U Rack Mount with no scale-up

# OAC Fit For Hyperscale Adjacent Systems

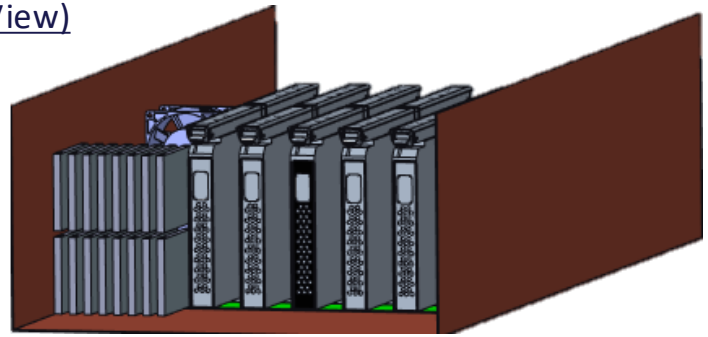
Scale-up be created in rack and stack fashion Off-The Shelf OAC



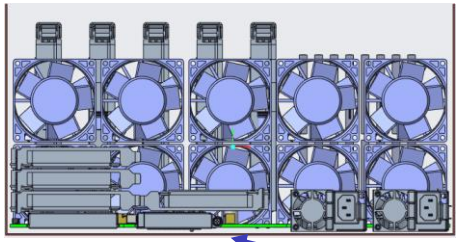
1000mm 5U
Enterprise Server with OAC
(Front View)

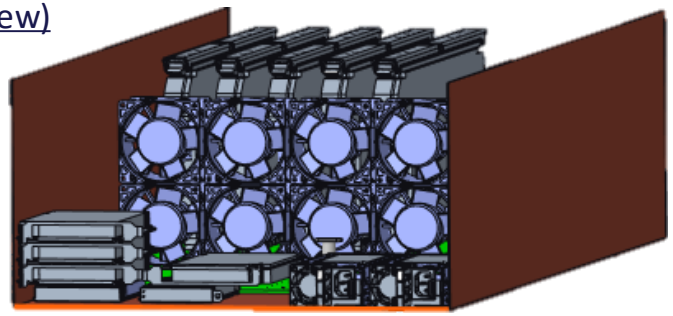



OAC Switch Card

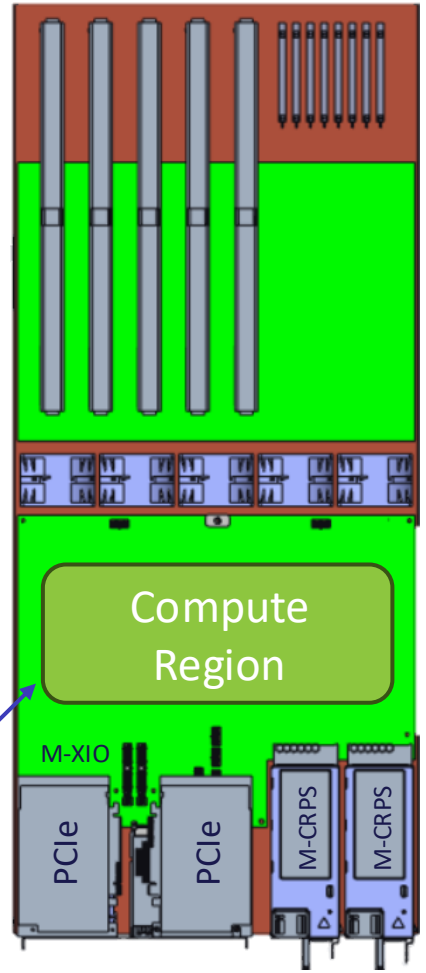


1000mm 5U
Enterprise Server with OAC
(Rear View)





OCP NIC SFF




OCP M-SDNO Class E

# Call to Action and Open Floor

## Need Community Involvement To Define Need and Solution

- If there is interest to investigate if a new form factor is required or driving system level specifications around OAC, OCP NIC, and PCIe please contact
  - ➢ dirk.Blevins@intel.com or vincent.nguyen@hpe.com
- Join our monthly AICC calls to drive the specifications – 3rd Tuesday of every month
- Subscribe to the AICC mailing list by sending an email to ocp-aicc+subscribe@OCP-All.groups.io with “Subscribe” in the subject line

# Thank You!

